4.

Right of A、B, 8 capacitors (Fig. 1) equal one C_{AB}. Simplify to Fig. 2. First find C_{AB}.

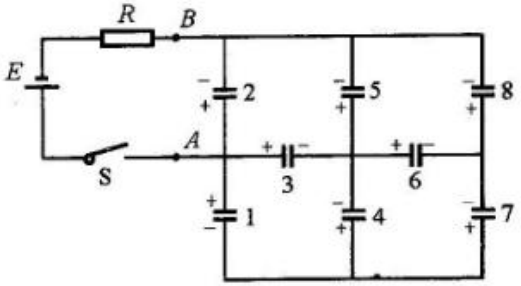


Fig. 1

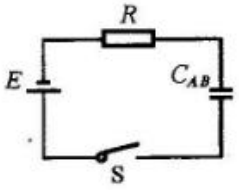


Fig. 2

S shut: final charge from A、B to right of A、B is Q (A positive, B negative). Capacitor i: q_i; plate charge signs are as in Fig. 1.

Node charge balance: 4 independent equations:

$$q_1 + q_2 + q_3 = Q \tag{1}$$

$$-q_1 + q_4 + q_7 = 0 \tag{2}$$

$$-q_3 - q_4 + q_5 + q_6 = 0 \tag{3}$$

$$-q_6 - q_7 + q_8 = 0 \tag{4}$$

Loop voltages, e.g. $\frac{q_1}{C} + \frac{q_4}{C} - \frac{q_3}{C} = 0$, give

$$q_1 + q_4 - q_3 = 0 \tag{5}$$

$$q_4 + q_6 - q_7 = 0 \tag{6}$$

$$q_2 - q_3 - q_5 = 0 \tag{7}$$

$$q_5 - q_6 - q_8 = 0 \tag{8}$$

Both cap. 2 and equivalent C_{AB} have voltage E; hence

$$E = \frac{q_2}{C} \tag{9}$$

$$E = \frac{Q}{C_{AB}} \tag{10}$$

Solving (1) ~ (8) gives

$$q_2 = \frac{8}{15}Q \tag{11}$$

From (9)、(10)、(11),

$$C_{AB} = \frac{15}{8}C \tag{12}$$

During charging in Fig. 2, total circuit charge is Q; the source does work

$$W_0 = EQ \tag{13}$$

Energy stored in the equivalent capacitor:

$$W_1 = \frac{1}{2}QE \tag{14}$$

Energy conservation: heat dissipated in the resistor is

$$W_2 = W_0 - W_1 \tag{15}$$